ORDINARY DIFFERENTIAL EQUATIONS • UNIT 6.01

C The Laplace Transform, Step Forcing & Convolution

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

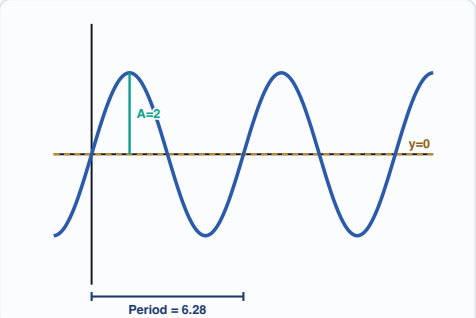
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 D	Q2 B	Q3 C	Q4 B	Q5 C
Q6 A	Q7 A	Q8 A	Q9 D	Q10 B
Q11 A	Q12 C	Q13 B	Q14 D	Q15 B
Q16 A	Q17 B	Q18 D	Q19 D	Q20 B
Q21 C	Q22 C	Q23 D	Q24 D	Q25 D

The Laplace Transform, Step Forcing & Convolution Practice Exercises

1 MCQ

Find $\mathrm{L}\{3t + \cos(t)\}$.



A $\frac{2s^3+3s^2+s+3}{s^4+s^2}$

B $\frac{-s^3-3s^2-3}{s^4+s^2}$

C $\frac{s^3+3s^2+3}{s^3+s}$

D $\frac{s^3+3s^2+3}{s^4+s^2}$ ✓

WORKED SOLUTION & VERIFICATION

Use the table (and linearity): $\mathrm{L}\{3t + \cos(t)\} = \frac{s^3+3s^2+3}{s^4+s^2}$.

2 MCQ

Using the definition, $\mathrm{L}\{e^{2t}\} = \int_0^\infty e^{-(s-2)t}\,dt$ equals (for $s > 2$):

A $\frac{s}{s-2}$

B $\frac{1}{s-2}$ ✓

C $\frac{1}{s+2}$

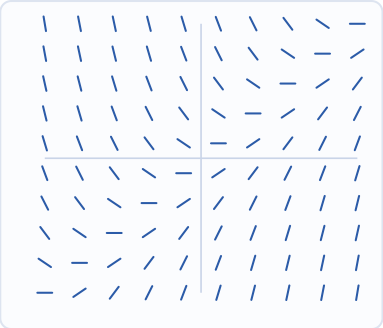
D $-\frac{1}{s-2}$

WORKED SOLUTION & VERIFICATION

Evaluate the improper integral; the boundary term at infinity vanishes for $s > 2$, leaving $\frac{1}{s-2}$.

3 MCQ

If $\mathrm{L}\{y\} = Y(s)$, $y(0) = -3$ and $y'(0) = -1$, find $\mathrm{L}\{y''\}$.



A $Y s^2 + s + 3$

B $Y s^2 - 3s - 1$

C $Y s^2 + 3s + 1$ ✓

D $Y s + 4$

WORKED SOLUTION & VERIFICATION

$\mathrm{L}\{y''\} = s^2 Y(s) - sy(0) - y'(0) = Y s^2 + 3s + 1.$

4 MCQ

Find $\mathrm{L}\{t^4\}$.

A $\frac{24}{s^4}$

B $\frac{24}{s^5}$ ✓

C $\frac{4}{s^5}$

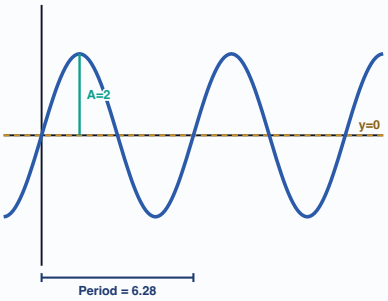
D $\frac{24}{s^6}$

WORKED SOLUTION & VERIFICATION

Use the table (and linearity): $\mathrm{L}\{t^4\} = \frac{24}{s^5}$.

5 MCQ

Using the definition, $\mathrm{L}\{2t\} = \int_0^\infty 2te^{-st}\,dt$ equals (for $s > 0$):



A $\frac{2}{s}$

B $\frac{2}{s^3}$

C $\frac{2}{s^2}$ ✓

D $-\frac{2}{s^2}$

WORKED SOLUTION & VERIFICATION

Evaluate the improper integral; the boundary term at infinity vanishes for $s > 0$, leaving $\frac{2}{s^2}$.

6 MCQ

The second-order linear homogeneous ODE $f''(x) - 53f'(x) + 696f(x) = 0$ has characteristic roots λ_1, λ_2. What is the trace sum $\lambda_1 + \lambda_2$?

- **A** 53 ✓
- **B** 56
- **C** 77
- **D** 28

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 53\lambda + 696 = 0$, the sum of roots is 53.

7 MCQ

If $\mathrm{L}\{y\} = Y(s)$ and $y(0) = 1$, find $\mathrm{L}\{y'\}$.

- **A** $Ys - 1$ ✓
- **B** $Ys + 1$
- **C** $Y - 1$
- **D** Ys

WORKED SOLUTION & VERIFICATION

$\mathrm{L}\{y'\} = sY(s) - y(0) = Ys - 1.$

8 MCQ

Find $\mathrm{L}\{t^3\}$.

- **A** $\frac{6}{s^4}$ ✓
- **B** $\frac{6}{s^3}$
- **C** $\frac{3}{s^4}$
- **D** $\frac{6}{s^5}$

WORKED SOLUTION & VERIFICATION

Use the table (and linearity): $\mathrm{L}\{t^3\} = \frac{6}{s^4}$.

9 MCQ

Consider the differential equation $y''(t) + 4y(t) = u(t-2)$, where $u(t)$ is the Heaviside step function, with initial conditions $y(0) = 0$ and $y'(0) = 0$. Apply the Laplace Transform to both sides of the equation. If $Y(s) = \mathrm{L}\{y(t)\}$ and $\mathrm{L}\{u(t-a)\} = \frac{e^{-as}}{s}$, which of the following expressions represents $Y(s)$?

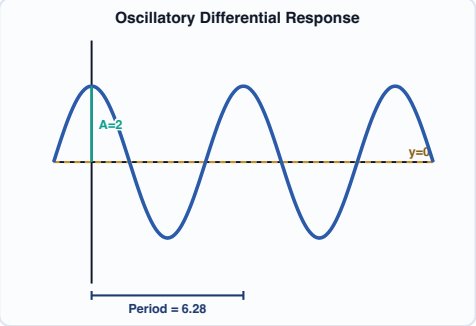
- **A** $\frac{1}{(s^2+4)}$
- **B** $\frac{1}{s(s^2+4)}$
- **C** $\frac{e^{-2s}}{s^2+4}$
- **D** $\frac{e^{-2s}}{s(s^2+4)}$ ✓

WORKED SOLUTION & VERIFICATION

Applying the Laplace Transform to the given differential equation: $\mathrm{L}\{y''(t)\} + 4\mathrm{L}\{y(t)\} = \mathrm{L}\{u(t-2)\}$. Using the properties of the Laplace Transform and the given initial conditions $y(0) = 0$ and $y'(0) = 0$, we have $\mathrm{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0) = s^2Y(s)$. The Laplace Transform of $y(t)$ is $Y(s)$. The Laplace Transform of the Heaviside step function $u(t-2)$ is $\frac{e^{-2s}}{s}$. Substituting these into the transformed equation gives: $s^2Y(s) + 4Y(s) = \frac{e^{-2s}}{s}$. Factoring out $Y(s)$, we get $Y(s)(s^2+4) = \frac{e^{-2s}}{s}$. Solving for $Y(s)$, we obtain $Y(s) = \frac{e^{-2s}}{s(s^2+4)}$.

10 MCQ

The second-order linear homogeneous ODE $f''(x) - 39f'(x) + 350f(x) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.



A 24

B 39 ✓

C 42

D 53

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 39\lambda + 350 = 0$, the sum of roots is 39.

11 MCQ

Consider the initial value problem $y'' + 4y = \delta(t - \pi)$ with $y(0) = 0$ and $y'(0) = 0$. Which of the following represents the Laplace Transform of the solution, $Y(s) = \mathrm{L}\{y(t)\}$?

A $\frac{e^{-\pi s}}{s^2+4}$ ✓

B $\frac{1}{s^2+4}$

C $\frac{e^{-\pi s}}{s^2}$

D $\frac{e^{-\pi s}}{s^2-4}$

WORKED SOLUTION & VERIFICATION

Taking the Laplace transform of the differential equation, we have $s^2Y(s) - sy(0) - y'(0) + 4Y(s) = \mathrm{L}\{\delta(t - \pi)\}$. Given the initial conditions $y(0) = 0$ and $y'(0) = 0$, this simplifies to $s^2Y(s) + 4Y(s) = e^{-\pi s}$. Factoring out $Y(s)$, we get $Y(s)(s^2 + 4) = e^{-\pi s}$. Therefore, the Laplace transform of the solution is $Y(s) = \frac{e^{-\pi s}}{s^2+4}$.

12 MCQ

The separable initial value problem $\frac{df}{dx} = 40x$ satisfies $f(0) = 4$. Determine the solution value $f(16)$.

A 5144

B 5120

C 5124 ✓

D 5127

WORKED SOLUTION & VERIFICATION

Integrating gives $f(x) = 20x^2 + 4$, so $f(16) = 5124$.

13 MCQ

A damped spring satisfies $y'' + 2y' + 10y = 0$, $y(0) = 0$, $y'(0) = 6$. Find $y(\pi/6)$.

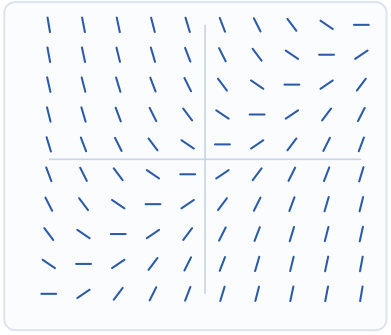
- **A** $2e^{\pi/6}$
- **B** $2e^{-\pi/6}$ ✓
- **C** $-2e^{-\pi/6}$
- **D** $e^{-\pi/6}$

WORKED SOLUTION & VERIFICATION

Auxiliary: $r^2 + 2r + 10 = 0 \Rightarrow r = \dfrac{-2 \pm \sqrt{4 - 40}}{2} = -1 \pm 3i - \alpha = -1, \beta = 3$, so $y = e^{-x}(C_1 \cos 3x + C_2 \sin 3x)$. $y(0) = C_1 = 0$, so $y = C_2 e^{-x} \sin 3x$; $y' = C_2 e^{-x}(3\cos 3x - \sin 3x)$, $y'(0) = 3C_2 = 6 \Rightarrow C_2 = 2$. So $y = 2e^{-x} \sin 3x$, and $y(\pi/6) = 2e^{-\pi/6} \sin(\pi/2) = 2e^{-\pi/6}$.

14 MCQ

Solve $y'' + 4y' + 4y = e^{-2x}$, $y(0) = 0$, $y'(0) = 0$. Find $y(2)$.



- **A** $2e^4$
- **B** e^{-4}
- **C** $4e^{-4}$
- **D** $2e^{-4}$ ✓

WORKED SOLUTION & VERIFICATION

Homogeneous: $r^2 + 4r + 4 = (r+2)^2 = 0 \Rightarrow r = -2$ (repeated), $y_c = (C_1 + C_2 x)e^{-2x}$. The forcing e^{-2x} matches the repeated root, so the trial needs x^2: $y_p = Ax^2 e^{-2x}$. Substituting gives $2Ae^{-2x} = e^{-2x} \Rightarrow A = \frac{1}{2}$, so $y_p = \frac{1}{2}x^2 e^{-2x}$. With both conditions zero, $C_1 = C_2 = 0$. So $y = \frac{1}{2}x^2 e^{-2x}$, and $y(2) = \frac{1}{2}(4)e^{-4} = 2e^{-4}$.

15 MCQ

Calculate the solution set for $y'' - 7y' + 12y = 0$, $y(0) = 0$, $y'(0) = 3$. Find $y(1)$.

- **A** $3e^3 + 3e^4$
- **B** $-3e^3 + 3e^4$ ✓
- **C** $-3e^4 + 3e^3$
- **D** $-6e^3 + 6e^4$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 - 7r + 12 = 0$ has roots $r = 4, 3$, so $y = C_1 e^{4x} + C_2 e^{3x}$. Solving $y(0) = C_1 + C_2 = 0$ and $y'(0) = 4C_1 + 3C_2 = 3$ gives $C_1 = 3$, $C_2 = -3$. So $y = 3e^{4x} - 3e^{3x}$, and $y(1)$ follows by substitution.

16 MCQ

Solve $y'' + y' - 2y = 0$, $y(0) = 0$, $y'(0) = -6$. Find $y(1)$.

- **A** $-2e + \frac{2}{e^2}$ ✓
- **B** $-4e + \frac{4}{e^2}$
- **C** $-2e - \frac{2}{e^2}$
- **D** $-\frac{2}{e^2} + 2e$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 + r - 2 = 0$ has roots $r = 1, -2$, so $y = C_1 e^x + C_2 e^{-2x}$. Solving $y(0) = C_1 + C_2 = 0$ and $y'(0) = C_1 - 2C_2 = -6$ gives $C_1 = -2$, $C_2 = 2$. So $y = -2e^x + 2e^{-2x}$, and $y(1)$ follows by substitution.

17 MCQ

Find the root(s) satisfying $y'' - 2y' + y = 0$, $y(0) = 1$, $y'(0) = 2$. Find $y(1)$.

- **A** 0
- **B** $2e$ ✓
- **C** $-2e$
- **D** $\frac{2}{e}$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 - 2r + 1 = 0 \Rightarrow (r-1)^2 = 0 \Rightarrow r = 1$ (repeated), so $y = (C_1 + C_2 x)e^x$. $y(0) = C_1 = 1$; $y'(0) = C_2 + C_1 = 2 \Rightarrow C_2 = 1$. So $y = (1 + x)e^x$, and $y(1)$ follows by substitution.

18 MCQ

Solve $y'' + 9y = 0$, $y(0) = -3$, $y'(0) = 6$. Find $y(\frac{\pi}{3})$.

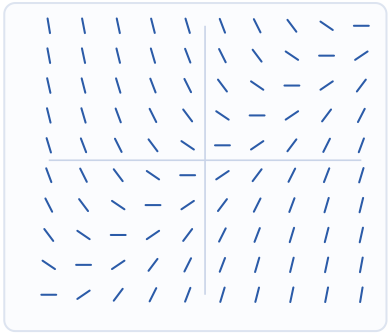
- **A** -2
- **B** -3
- **C** 6
- **D** 3 ✓

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 + 9 = 0$ has roots $r = 0 \pm 3i$, so $y = C_1 \cos 3x + C_2 \sin 3x$. $y(0) = C_1 = -3$; $y'(0) = 3C_2 = 6 \Rightarrow C_2 = 2$. Evaluating at $x = \frac{\pi}{3}$ (where $3x = \pi$) gives the result.

19 MCQ

Calculate the solution set for $y'' + 4y = 0$, $y(0) = -2$, $y'(0) = 6$. Find $y(\frac{\pi}{2})$.



A 4

B -3

C -2

D 2 ✓

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 + 4 = 0$ has roots $r = 0 \pm 2i$, so $y = C_1 \cos 2x + C_2 \sin 2x$. $y(0) = C_1 = -2$; $y'(0) = 2C_2 = 6 \Rightarrow C_2 = 3$. Evaluating at $x = \frac{\pi}{2}$ (where $2x = \pi$) gives the result.

20 MCQ

Solve $y'' - 4y' + 3y = 0$, $y(0) = -5$, $y'(0) = -7$. Find $y(1)$.

A $-4e^3 - e$

B $-e^3 - 4e$ ✓

C $4e + e^3$

D $-4e + e^3$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 - 4r + 3 = 0$ has roots $r = 1, 3$, so $y = C_1 e^x + C_2 e^{3x}$. Solving $y(0) = C_1 + C_2 = -5$ and $y'(0) = C_1 + 3C_2 = -7$ gives $C_1 = -4$, $C_2 = -1$. So $y = -4e^x - e^{3x}$, and $y(1)$ follows by substitution.

21 MCQ

Find the root(s) satisfying $y'' + 3y' + 2y = 0$, $y(0) = 5$, $y'(0) = -7$. Find $y(2)$.

A $-\frac{3}{e^2} - \frac{2}{e^4}$

B $-\frac{3}{e^2} + \frac{2}{e^4}$

C $\frac{2}{e^4} + \frac{3}{e^2}$ ✓

D $\frac{3}{e^4} + \frac{2}{e^2}$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 + 3r + 2 = 0$ has roots $r = -2, -1$, so $y = C_1 e^{-2x} + C_2 e^{-x}$. Solving $y(0) = C_1 + C_2 = 5$ and $y'(0) = -2C_1 - C_2 = -7$ gives $C_1 = 2$, $C_2 = 3$. So $y = 2e^{-2x} + 3e^{-x}$, and $y(2)$ follows by substitution.

22 MCQ

Determine the solution to $y'' - 6y' + 9y = 0$, $y(0) = 2$, $y'(0) = 8$. Find $y(2)$.

A $-6e^6$

B $\frac{6}{e^6}$

C $6e^6$ ✓

D $-2e^6$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 - 6r + 9 = 0 \Rightarrow (r-3)^2 = 0 \Rightarrow r = 3$ (repeated), so $y = (C_1 + C_2 x)e^{3x}$. $y(0) = C_1 = 2$; $y'(0) = C_2 + 3C_1 = 8 \Rightarrow C_2 = 2$. So $y = (2 + 2x)e^{3x}$, and $y(2)$ follows by substitution.

23 MCQ

Work out the solution to $y'' + 6y' + 9y = 0$, $y(0) = -2$, $y'(0) = 3$. Find $y(1)$.

A e^{-3}

B $\frac{5}{e^3}$

C $-5e^3$

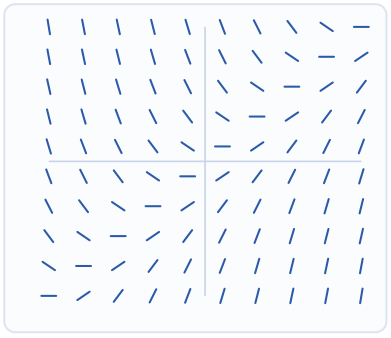
D $-\frac{5}{e^3}$ ✓

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 + 6r + 9 = 0 \Rightarrow (r+3)^2 = 0 \Rightarrow r = -3$ (repeated), so $y = (C_1 + C_2 x)e^{-3x}$. $y(0) = C_1 = -2$; $y'(0) = C_2 - 3C_1 = 3 \Rightarrow C_2 = -3$. So $y = (-2 - 3x)e^{-3x}$, and $y(1)$ follows by substitution.

24 MCQ

Solve $y'' - 4y' + 8y = 0$, $y(0) = -1$, $y'(0) = 2$. Find $y(\frac{\pi}{6})$.



A $-\sqrt{3}e^{\frac{\pi}{3}} - \frac{e^{\frac{\pi}{3}}}{2}$

B $-\frac{\sqrt{3}e^{\frac{\pi}{3}}}{2} + e^{\frac{\pi}{3}}$

C $-\sqrt{3}e^{\frac{\pi}{3}} + \frac{e^{\frac{\pi}{3}}}{2}$

D $-\frac{e^{\frac{\pi}{3}}}{2} + \sqrt{3}e^{\frac{\pi}{3}}$ ✓

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 - 4r + 8 = 0$ has roots $r = 2 \pm 2i$, so $y = e^{2x}(C_1 \cos 2x + C_2 \sin 2x)$. $y(0) = C_1 = -1$; $y'(0) = 2C_1 + 2C_2 = 2 \Rightarrow C_2 = 2$. Evaluating at $x = \frac{\pi}{6}$ (where $2x = \frac{\pi}{3}$) gives the result.

25 MCQ

Work out the solution to $y'' - 12y' + 35y = 0$, $y(0) = -1$, $y'(0) = -13$. Find $y(2)$.

A $3e^{10} + 4e^{14}$

B $-4e^{10} + 3e^{14}$

C $-3e^{10} + 4e^{14}$

D $-4e^{14} + 3e^{10}$ ✓

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 - 12r + 35 = 0$ has roots $r = 5, 7$, so $y = C_1e^{5x} + C_2e^{7x}$. Solving $y(0) = C_1 + C_2 = -1$ and $y'(0) = 5C_1 + 7C_2 = -13$ gives $C_1 = 3$, $C_2 = -4$. So $y = 3e^{5x} - 4e^{7x}$, and $y(2)$ follows by substitution.